UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

5090 BIOLOGY

5090/03 Paper 3 (Practical Test), maximum raw mark 40

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1 (a) (i) all 20 boxes reasonably filled, with units somewhere in box ; [1]

(ii) table marks:
all boxes numbered, correct regular sequence ;
clear method of 'tally', somewhere ;
third column grouping correct, totalling 20 ; [3]

(iii) frequency diagram marks ;
ruled, regular columns ;
x-axis labelled, 'length/mm/cm' ;
y-axis labelled 'number (frequency)/& numbered. ('o' stated or implied) ;
range of each column centrally stated on *x*-axis ;
accurate for numbers in table ;
axes reversed: max 3 [5]

(b) seeds of different sizes ;
grown in uniform conditions ;
two of them specified ;
(soil/medium, temp, light, water, fertiliser – nutrient) ;
seeds (collected and) measured ;
recorded, (tabulated, frequency chart) from each plant ;
replications ;
analysis/conclusion suggested ;

A: ref same size seeds ; in different conditions ;
OVP ; [max 4]

(c) (i) clear, clean realistic selection for drawing ;
radicle + plumule at least 5 cm (in straight line) ;
leaflets/venation shown on plumule ;
labels: radicle and plumule ;
label: leaflet/venation/attachment scar or other detail ; [5]

(ii) clearly shows where measured with both dimensions clear ;
units at least once ;
correct expression for calculation ;
magnification correct and realistic ; [max 3]

up to 2 d.p. rounded up/down acceptably.

(d) (i) mauve (etc)/ref. different from control drops ; [1]

(ii) ref. protein presence ; [1]

(iii) ref. growing regions/for growth ;
cells present etc ;
ref. enzyme (release here) ; [max 2]

[Total: 25]

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2 (a) (i) chop/crush into small bits ;
add ethanol & shake (to form solution) ;
add drops of water/pour into water ; [3]

(ii)

	observations	conclusions
W3	remains (relatively) clear no change/colourless ; R: 'negative'	no fat/oil ; *
W4	goes cloudy/white/milky ;	Fat/oil present ; *

[4]

(* if conclusions ' -ve & +ve' allow 1)

(b) hold piece of specimen on mounted needle/forceps – ignite it ;
known mass of specimen ;
heat (standard volume) water in test tube ;
ensure complete combustion ;
record increase in temp. ;
repeat for other specimen ;
same volume of water ;
one that burns longest ;
replicate ; [max 5]

(c) each has same amount energy from starch ;
fat (oil) content differs/**W4** has more fat/oil than **W3** ;
and yields energy/fat has high energy ;
W4 has more energy ; [max 3]

[Total: 15]

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